

AAF

Addressed Admissible Futures Future-Sufficient State, Load-Bearing Relations, and Restorative Reachability

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Publication boundary. This paper is the standalone AAF v0.1 publication object. It consolidates the Addressed Admissible Futures formulation from the live MKUFT canon without replacing the evolving module owners. A later DOI-bearing deposit fixes the exact deposited file for that version; later repository revisions are not silently inserted into that frozen object.

Abstract

Many dynamical systems are described by their present state, yet the variables required to predict lawful continuation may include momentum, phase, retained history, boundary conditions, relational context, or other coordinates not visible in an instantaneous snapshot. This paper formalises that distinction by introducing the Addressed Admissible Future (AAF): the future-equivalence object generated by a state description that is sufficient for a declared target, horizon, intervention class and tolerance. The central move is operational. An Address is not defined as a list of preferred variables. It is defined by predictive closure: two histories share an effective Address only when they induce the same admissible future law, exactly or within a declared tolerance. This immediately yields a minimum-sufficient address frontier, an upward-closed family of sufficient descriptions whose minimal elements need not lie on one scale.

The second move concerns recovery. Present viability and future recoverability are treated as distinct dynamical objects. Within the admissible future, the restorative subset contains trajectories that preserve load-bearing viability constraints and reach a declared restored target without unnecessary irreversible deformation. A restorative-future reserve measures how much relation structure must be removed or deformed before every sufficient restorative continuation is lost. This turns “load-bearing relation” into a counterfactual definition: a relation is load-bearing relative to a declared functional future when removing it materially deforms that future, even if the relation carries little or no present flow.

Three formal consequences follow. First, a currently viable system may already have a fragile or empty restorative future. Second, variables such as velocity, phase or hysteresis enter Address only when omitting them splits the future. Third, traversal is recursively readdressing: a realised transition changes the history and therefore may change the future-equivalence class governing the next transition.

The framework is anchored to established mathematics in predictive state representations, computational mechanics, viability theory, control-barrier methods, graph cuts and, where recurrent flow itself is empirically load-bearing, Hodge decomposition. A bounded braking example shows the separation between present position and future-sufficient state. Biological calibration is supplied by single-cell work showing that p21/Cdk2 position plus velocity predicts fate better than position alone, and that p53 phase resetting changes downstream p21 dynamics and cell-cycle arrest robustness. The paper ends with a preregistered test using public p53/p21 single-cell data. No empirical result is claimed here. The contribution is a falsifiable formal object and a fixed route for testing whether future structure carries predictive information beyond conventional present-state summaries.

Keywords: addressed state; predictive sufficiency; admissible future; viability; recoverability; load-bearing relation; minimum-sufficient address frontier; restorative trajectory; state history; p53; p21; dynamical systems.

Status and claim boundary

This is a standalone research paper prepared for public dissemination and empirical testing. It develops a formal object from the current MKUFT and TVT architecture. It does not claim a new fundamental force, a new biological mechanism, or a replacement for viability theory, predictive-state theory, control theory, graph theory or dynamical-systems biology. Established mathematics is recruited where the object requires it. The paper succeeds only to the extent that the formalism produces discriminating, prospective, held-out results under strong controls.

The current principal MKUFT publication defines complete addressed state, retained load-bearing history, admissibility, readdressing, property-relative law sufficiency and target-relative recoverability as distinct objects [1]. TVT adds an ordered safety architecture: preserve the evidence state, type the address, identify legitimate nested viability, prefer sufficient repair/restoration before irreversible escalation, filter gratuitous irreversible loss, choose among the survivors, then re-sense and re-address [2]. The present paper asks what mathematics is forced when the future structure produced by that sequence is itself load-bearing.

Framework placement. AAF is not a separate law stack beside MKUFT. It is a future-object specialisation of the existing recursive S-I-P-O addressed-update architecture: addressed state $\rightarrow$ admissibility $\rightarrow$ lawful continuation $\rightarrow$ realised transition/registration $\rightarrow$ readdressing. AAF asks which future-equivalence object is induced by a state description sufficient for the declared target, and then measures restorative reachability, reserve and load-bearing relation structure inside that future. The scientific burden therefore remains with the existing MKUFT rule: a new coordinate or relation earns Address status only when it changes prospective predictive or interventional closure beyond the strongest adequate baseline.

1. The problem: a snapshot is not necessarily a state

Consider two objects occupying the same measured position at the same clock time. If one has positive momentum and the other negative momentum, their lawful continuations differ. Position alone is therefore not a complete state description for the target “where can this object be next?” The same logic applies more generally. In a driven oscillator,

phase can split future response. In a hysteretic material, identical present observables can conceal different retained histories. In a biological network, cells with similar instantaneous marker levels can still diverge because velocity, chromatin state, damage history or coupling differs.

The usual response is to add more variables. That is necessary but incomplete because it leaves open a prior question: which variables must be retained, and when is the description sufficient?

AAF answers that question by defining Address through the future rather than through a fixed ontology of coordinates.

Truth remains prior. Let e be the current evidential state, o a new observation and p a preferred outcome. A factual update operator B satisfies epistemic non-interference when changing the preferred outcome does not silently change the update generated from the same evidence:

$$B(e, o; p_1) = B(e, o; p_2) \quad \text{for fixed evidence } (e, o).$$

This is not a complete theory of inference. It fixes the boundary required for the later construction: the future must be conditioned on the best available state estimate, not on a desired answer.

2. Future-sufficient Address

2.1 Exact future equivalence

Let H_t denote the retained history available at time t . Let Σ denote the declared system specification: target variables, horizon, physical dynamics or transition kernel, boundary and environment, admissible intervention class, measurement resolution, and any typed relational or authority constraints relevant to the task.

Let X^+ denote the future path object after t . In a controlled stochastic system, two histories are future-equivalent when they generate the same future law for every admissible policy:

$$H_t \sim_{\Sigma} H'_t \iff \mathbb{P}_{\Sigma}(X^+ \mid H_t, \pi) = \mathbb{P}_{\Sigma}(X^+ \mid H'_t, \pi) \quad \forall \pi \in \Pi_{\text{adm}}.$$

The effective Address is the equivalence class:

$$A_t = [H_t]_{\sim_{\Sigma}}.$$

This definition is deliberately severe. It says that an Address is complete only when histories collapsed into the same Address no longer split the admissible future under the declared construction.

In deterministic systems, equality of future probability laws reduces to equality of the reachable/admissible future set. In noisy empirical systems, exact equality is replaced by a declared discrepancy and tolerance.

2.2 Operational target-relative address

Full future equivalence is usually impossible to estimate from finite data. For an operational target q over horizon Δ , let $\Theta(H_t)$ be a candidate coordinate description extracted from history. Define the target-relative address residual

$$\mathcal{R}_q(\Theta) = \sup_{\Theta(h)=\Theta(h')} d(\mathcal{L}(q^+ \mid h), \mathcal{L}(q^+ \mid h')).$$

where $\mathcal{L}(q^+ \mid h)$ is the conditional law of the declared future target and d is a preregistered discrepancy. Θ is ε_q -sufficient when $\mathcal{R}_q(\Theta) \leq \varepsilon_q$.

This residual gives a direct falsifier for address completeness. If an omitted variable y partitions a Θ -class into subgroups with materially different future target laws, then Θ was incomplete for q under the declared regime.

2.3 Monotone refinement

Suppose Θ_2 lawfully augments Θ_1 , meaning Θ_1 is retained and Θ_2 adds coordinates without dropping load-bearing parent context. Then the equivalence classes induced by Θ_2 refine those induced by Θ_1 . Therefore

$$\Theta_1 \subseteq \Theta_2 \implies \mathcal{R}_q(\Theta_2) \leq \mathcal{R}_q(\Theta_1).$$

because the supremum is taken over a smaller set of same-address history pairs.

This yields an immediate order-theoretic result. Define

$$\mathcal{S}_q = \{\Theta : \mathcal{R}_q(\Theta) \leq \varepsilon_q\}, \quad \mathcal{M}_q = \{\Theta \in \mathcal{S}_q : \text{no proper sufficient subset}\}.$$

The sufficient family $\mathcal{S}_q$ is upward closed under lawful augmentation, while its minimally sufficient members $\mathcal{M}_q$ form an antichain: distinct minimal sufficient descriptions need not contain one another. There may therefore be several incomparable minimal predictive handles for the same declared future.

This is the minimum-sufficient address frontier. It replaces the question “what is the one correct scale?” with a testable question: what is the shallowest complete description, or family of incomparable complete descriptions, for this target now?

The frontier is target-, horizon-, intervention-, environment-, resolution- and tolerance-relative. A higher-resolution description is not automatically better if it discards a load-bearing higher-level boundary or relation. Adding microscopic detail while dropping tissue context, for example, is a different address rather than a lawful refinement.

3. Admissible future

The Address determines a family of future continuations. Let $F+(A_t)$ denote the admissible future from Address A_t under the declared system specification. It can be represented as a reachable set, a family of path measures, a directed state-transition structure, or another domain-appropriate carrier.

The important distinction is between the complete future and properties measured on that future. Topology, flow, recurrence, path count, energy cost and restoration probability are possible readouts. None is automatically the parent object.

For physical systems, “admissible” means compatible with the governing dynamics, boundary conditions and intervention constraints. For a typed decision architecture, it also includes provenance, authority and consequence constraints when those genuinely determine which transitions may lawfully be taken.

4. Present viability

Let $J(a)$ index the load-bearing viability constraints at address a . Define

$$\mathcal{V}(a) = \{z : h_j(z; a) \geq 0 \ \forall j \in J(a)\}, \quad H(z, a) = \min_{j \in J(a)} h_j(z; a).$$

The scalar H is a present-state margin: it reports the smallest declared viability margin among the currently load-bearing constraints. $H > 0$ means the state is presently inside the viable region, $H = 0$ is the boundary, and $H < 0$ is outside.

This object is intentionally local in time. It must not be confused with recoverability. A system can be safe now while its routes for remaining safe or returning to a restored condition have already narrowed.

5. The restorative future

Let $G(A_t)$ be a declared restored or recovered target set. Let $Q(\gamma)$ measure whether a trajectory γ is sufficient for the actual task, with minimum sufficiency κ . Let $D_{\text{irr}}(\gamma)$ be a vector of domain-appropriate irreversible losses with declared upper bounds d_R .

Define the restorative future

$$\Gamma_R(A_t) = \{\gamma : \gamma \subseteq \mathcal{V}(A_t), \gamma \rightarrow G(A_t), \\ Q(\gamma) \geq \kappa, D_{\text{irr}}(\gamma) \preceq d_R\}.$$

The definition is typed rather than sentimental. “Restorative” means that the path stays within required viability constraints, reaches the declared target, actually solves enough of the problem, and stays within the allowed irreversible-loss boundary.

The simplest existence indicator is

$$\chi_R(A_t) = \mathbf{1}[\Gamma_R(A_t) \neq \emptyset].$$

This separates three conditions that ordinary language often compresses:

- (1) presently viable;
- (2) at least one sufficient restorative future remains;
- (3) the remaining restorative future is robust rather than knife-edge.

6. Restorative-future reserve

Existence alone is brittle. A system can possess one technically valid repair path that disappears after a negligible perturbation. Define a family $\mathcal{D}$ of admissibility-structure deformations, with domain-defined cost $c(\Delta)$. Then define restorative-future reserve as

$$\rho_R(A_t) = \inf_{\Delta \in \mathcal{D}} \{c(\Delta) : \Gamma_R(A_t; \Delta) = \emptyset\}.$$

High ρ_R means that a substantial deformation is required to eliminate every sufficient restorative continuation. Low ρ_R means that restoration technically remains possible but is structurally fragile. $\chi_R=0$ means no sufficient restorative future remains under the declared construction.

The deformation class and its cost cannot be selected after the outcome is known. In a graph approximation, Δ may remove or weaken directed transition edges. In a control system it may shrink actuation authority or move a boundary. In biology it may correspond to loss of a regulatory interaction, timing window or measured transition support. Different domains require different physically meaningful perturbation models.

7. Load-bearing relation

Let C be a relation or coalition of relations in the directed typed structure carrying the restorative future. Let Ψ be a preregistered functional readout of that future: existence, reserve, recovery probability, time-to-restoration, irreversible cost, or another declared quantity.

Define relational load as

$$\Lambda_C^\Psi(A_t) = \Psi(\Gamma_R(A_t)) - \Psi(\Gamma_R^{-C}(A_t)).$$

A relation is load-bearing relative to (A_t, Ψ) when its removal materially deforms the declared future readout. It is terminally load-bearing when removal changes χ_R from 1 to 0.

This definition rejects several weak proxies. High graph centrality is not enough. High observed flow is not enough. Repetition is not enough. A dormant backup relation may carry almost no current flow yet dominate future recoverability. Conversely, a heavily used relation may be replaceable by several alternatives and therefore carry little terminal load.

Coalitions are essential. Two relations can each appear dispensable under single-ablation while their joint removal destroys every restorative future. The object is therefore the deformation response of the future, not the visual prominence of an edge.

8. Readdressing as recursion

A realised transition does not merely move the object through a fixed possibility map. It changes the retained history and can therefore change the future-equivalence class that defines the next Address:

$$A_t \rightarrow \mathcal{F}_t^+ \rightarrow \tau_t \rightarrow A_{t+1} \rightarrow \mathcal{F}_{t+1}^+.$$

This is the recursive core. Traversal changes state; state changes Address; Address changes admissibility; admissibility changes the next future.

A variable such as timing, charge, velocity, phase, rate or hysteresis belongs in Address only when changing it while holding the coarser description fixed changes the admissible future beyond tolerance. This gives an operational membership rule:

A coordinate is address-load-bearing if omitting it leaves histories classified as the same Address even though their declared future laws materially differ.

9. Graph approximation and minimum cut

For finite data, approximate the restorative future by a directed graph or time-expanded transition network $\mathcal{K}_R$. Nodes represent addressed states or microstates; directed edges represent empirically supported admissible transitions. Let c_e be a preregistered edge capacity or robustness weight.

When the target is a set G , a graph estimator of restorative reserve is the minimum directed cut separating the present node s from all restorative target nodes:

$$\hat{\rho}_R = \min_{C \in \text{Cut}(s, G)} \sum_{e \in C} c_e.$$

By max-flow/min-cut duality, this can also be interpreted as the supported capacity of independent route structure from the present state to the restored target [12].

Raw path count is not a substitute. A thousand nominal routes that all pass through one fragile bottleneck can have less reserve than three genuinely independent routes.

10. Worked physical example: position is not a complete Address

Consider a one-dimensional body with bounded acceleration:

$$\dot{x} = v, \quad \dot{v} = u, \quad |u| \leq u_{\max}.$$

Let the current position satisfy $x < 1$, with $x = 1$ an unsafe boundary. Suppose “restoration” means stopping before crossing that boundary. The maximum braking magnitude is $u_{\max}$.

For positive velocity v , the minimum stopping distance is

$$d_{\text{stop}} = \frac{v^2}{2u_{\max}}, \quad \text{recoverable} \iff \frac{v^2}{2u_{\max}} \leq 1 - x.$$

Two objects can therefore share the same present position x and the same present viability margin $H = 1 - x$ while possessing different futures. If one has small v it can stop; if another has sufficiently large v it cannot, despite being equally far from the boundary at the snapshot.

This demonstrates three points without invoking any exotic physics.

First, present viability does not determine restorative future.

Second, velocity/momentum is forced into the sufficient Address because omitting it merges states with different admissible continuations.

Third, a system can satisfy

$$H_t > 0 \quad \text{while} \quad \rho_R(A_t) \downarrow 0.$$

when the margin to the present boundary remains positive while restorative reserve collapses.

11. Minimum-sufficient Address frontier as a moving object

The frontier $\mathcal{M}_q$ need not be static. Let λ be a control or physical-state parameter. Then $\mathcal{M}_q(\lambda, \Delta, U, E)$ can move as the system changes.

A coarse address can initially close the future target. Later, two states with the same coarse description may diverge because previously hidden lower-scale or cross-scale coordinates have become dynamically load-bearing. The coarse address residual $\mathcal{R}_q$ rises above tolerance, forcing readdressing.

This yields a direct experimental programme:

- identify an address that closes a future target at baseline;
- perturb the system;
- test whether the same address continues to close the target;
- add the smallest coordinate that restores closure;
- ablate or intervene on that coordinate;
- determine whether the frontier motion predicts the overt transition before conventional state failure.

The aim is not infinite microscopic descent. Stop when predictive/interventional closure is earned. Further zoom that does not change the target, boundary, falsifier, implementation or decision is not progress.

12. BIOLOGICAL CALIBRATION I: p21/CDK2 STATE VELOCITY

Venkatachalapathy et al. studied the p21/Cdk2-dependent quiescence–proliferation decision under hypoxia using single-cell trajectories on an energy landscape [8]. They found that initial position alone did not adequately explain fate heterogeneity; position plus cell-state velocity did substantially better. The authors interpret velocity as carrying

additional state information not visible in the instantaneous p21/Cdk2 position, potentially including epigenetic or other biochemical variables.

AAF does not claim this result as its own. It uses the result as a direct calibration of the Address rule. If two cells have similar coarse position but different future distributions because their velocities differ, then position is an incomplete Address for that target. Velocity earns inclusion because it splits the future.

The important methodological point is broader: “state” should not be declared complete by naming measured coordinates. Completeness is earned when same-address histories cease to split the target-relevant future.

13. BIOLOGICAL CALIBRATION II: p53 PHASE RESETTING

Venkatachalapathy et al. later showed that p53 oscillations can be phase-reset by repeated DNA-damage stimuli and that stimulus timing modulates downstream p21 dynamics and cell-cycle arrest robustness [9]. Two NCS schedules, including 4.0-hour and 5.5-hour intervals, produced different p21 timing and different patterns of escape from high-p21 arrest. Associated imaging data, gene-expression data, modelling code and analysis code are publicly deposited [10].

Again, the important object is not merely “oscillations matter.” The result demonstrates that temporal relation can carry future-discriminating information. In AAF terms, phase/schedule belongs in the effective Address when omitting it collapses histories that lead to materially different future target laws.

14. What the whirlpool does — and does not — mean

If observed transition flux f is defined on a suitable simplicial complex, discrete Hodge decomposition may separate gradient, local curl and harmonic components:

$$Y \not\vdash R_t \mid B_t, \quad R_t = (\chi_R, \rho_R).$$

This mathematics is established and is already used in single-cell trajectory inference; PHLOWER, for example, leverages the harmonic component of a Hodge-Laplacian construction to infer branching differentiation trajectories [11].

AAF does not require recurrence or circulation. An acyclic repair network can instantiate the entire framework, and a dormant backup route may be load-bearing despite carrying zero present flux. Hodge components are therefore optional diagnostics. They earn a place only if they add held-out predictive information beyond future existence, reserve and conventional trajectory summaries.

This prevents a useful visual or dynamical rhyme from being promoted into mechanism merely because it is attractive.

15. Strong predictions and failure conditions

15.1 Future structure adds information beyond present condition

Let B_t denote a preregistered baseline containing conventional present-state and history summaries. Let $R_t = (\chi_R, \rho_R)$. The strongest practical prediction is

$$f = f_G + f_C + f_H.$$

for at least some systems in which recovery routes degrade before overt state failure.

If R_t adds no held-out information after adequate baseline state/history variables are included, restorative-future structure has not earned additional load in that system.

15.2 Present viability can outlive recoverability

A system may remain inside its conventional viable region while ρ_R falls toward zero. If reserve only collapses at or after the conventional failure marker, the proposed early structural warning fails for that target.

15.3 History can be address-load-bearing

Matched instantaneous states can have different futures when their phase, velocity or hysteretic history differs. If a current-state-only model performs indistinguishably from a history-complete model, history is not load-bearing for that target under those conditions.

15.4 Relation structure must beat crude graph statistics

Degree-preserving or time-layer-preserving rewiring should degrade reserve prediction when the real directed relation structure is genuinely load-bearing. If rewired controls perform as well, the measured effect is probably generic graph density or sample support rather than meaningful relational geometry.

15.5 Bottleneck reserve must beat raw route count where bottlenecks exist

If raw path count performs as well as minimum-cut reserve across systems specifically constructed or selected to contain shared bottlenecks, the reserve formalism is not earning its additional machinery.

15.6 Address variables must survive ablation/restoration

If coordinate y is claimed to restore predictive closure, removing it should reopen the target-relative address residual on held-out data. Restoring y should re-close the target. If the claim is causal rather than predictive, independent intervention on y must alter the future in the predicted direction.

15.7 Cross-domain transfer cannot rewrite the object

A general claim contracts if each new system requires a different definition of Address, restoration, reserve or load-bearing relation chosen after the outcome is known. Domain-specific measurements may differ; the formal role must remain fixed.

16. PREREGISTERED FIRST TEST: p21 ARREST-MAINTENANCE FUTURE

16.1 Dataset

Primary test material will use the public single-cell p21-mVenus data associated with the 2025 p53 phase-resetting study [9,10]. The relevant cohorts include two-dose NCS schedules spaced 4.0 h and 5.5 h apart, with single-cell p21 measurements and multiple biological replicates. The published gross treatment-direction result is already known and is therefore not scored as a discovery result.

The empirical test is narrower: can a future-sufficient, directed restorative-reserve construction predict individual arrest escape risk beyond conventional summaries of the same pre-outcome p21 history?

16.2 Frozen prediction time and outcome

Prediction time: $t^* = 5.0$ h.

No measurement after 5.0 h may enter the predictor state.

Outcome $Y = 1$ if the cell undergoes the study-defined escape from high-p21 arrest during 5.0–8.0 h; $Y = 0$ if arrest is maintained through 8.0 h. The event definition and preprocessing route are inherited from the published analysis rather than chosen after inspecting prediction errors.

16.3 Candidate Address

At $t^* = 5.0$ h, the primary candidate Address contains:

- p21 at 2.0, 2.5, 3.0, 3.5, 4.0, 4.5 and 5.0 h;
- NCS regimen;
- elapsed time since the most recent dose.

The staggered p21 history is retained because the formal object predicts that current level alone may be insufficient when velocity/phase/history split the future. Alternative window lengths may be evaluated only as preregistered sensitivity analyses, not selected by outcome performance.

16.4 Directed transition construction

Use leave-one-biological-replicate-out evaluation.

Within each training fold:

1. cluster candidate Address vectors into a fixed number K of microstates;
2. construct time-addressed nodes (microstate, time);
3. add only observed forward transitions between adjacent measurement times;

4. estimate edge support from training cells;
5. bootstrap training cells to estimate edge stability;
6. define edge capacity as empirical transition support multiplied by bootstrap stability;
7. define the restorative target as the set of 8 h microstates satisfying the inherited non-escape/high-p21 criterion.

The held-out replicate is mapped into the training construction without using its future label.

16.5 Primary AAF readouts

For each held-out cell at 5 h:

- χ_R : whether any supported route to the 8 h arrest-maintenance target exists;
- ρ_R : directed minimum-cut capacity from the cell's present node to the restorative target;
- optional secondary summaries: number of edge-disjoint routes, shortest restorative time, and frontier sensitivity to removal of individual high-load edges.

16.6 Baselines

B0: regimen and elapsed time since dose.

B1: B0 plus instantaneous p21 at 5 h.

B2: B1 plus one-hour slope, three-hour mean and three-hour variance.

B3: regularized classifier using the entire same seven-point p21 history plus regimen/timing.

B4: B3 plus χ_R and ρ_R .

B3 is the crucial control. If an ordinary regularized predictor using the same information performs as well as the future-structure model, topology or reserve must not be credited merely because the representation is elegant.

16.7 Primary score and hit rule

Primary score: pooled out-of-fold log loss.

Secondary: area under the precision–recall curve, Brier score and calibration slope.

A primary HIT requires:

1. lower ρ_R predicts greater escape risk in the same direction in at least two held-out biological replicates;
2. B4 reduces pooled held-out log loss by at least 5% relative to B3;
3. a cell-level bootstrap 95% interval for the log-loss improvement excludes zero;
4. the real directed relation structure outperforms the rewired structural control.

Failure of any condition prevents the strong claim. Results may be reported as ambiguous or reduced rather than rescued by changing the construction after inspection.

16.8 Structural controls

Time-history control: scramble the order of pre-5 h history blocks within cells while preserving their marginal values.

Relation control: degree- and time-layer-preserving rewiring that preserves node count, edge count and approximate degree while destroying the learned transition relation.

Label control: permute escape labels within biological replicate and treatment strata.

Current-state control: remove lagged history and retain only $p21(5\text{ h})$, regimen and elapsed dose time.

If rewired relation structure performs as well as the real graph, the claimed load-bearing transition geometry fails this test. If current state performs as well as history, history is not address-load-bearing here. If prediction requires post-5 h data, the early-future claim fails.

16.9 Hodge analysis is secondary

No Hodge feature enters the primary model. Only after the primary result is frozen and scored will cyclic/harmonic flow be tested as an incremental predictor. It remains descriptive unless it adds reproducible held-out information beyond B4.

17. Relation to established mathematics

The Address construction intentionally meets several established bodies of work.

Computational mechanics groups histories into causal states according to equality of conditional future distributions and establishes predictive sufficiency/minimality properties [3]. Predictive state representations similarly represent dynamical state through predictions of future observations under action [4]. Bisimulation metrics formalise behavioural similarity of states in controlled stochastic systems [5]. AAF recruits that future-equivalence logic and embeds it inside a typed target/horizon/intervention construction with explicit address residuals and readdressing.

Viability theory asks which states admit trajectories that remain inside constraints and defines capture/reachability objects for targets [6]. Control-barrier methods formalise forward invariance and safety constraints [7]. AAF recruits these to distinguish present viability from the restorative subset of the admissible future.

Network cut theory supplies an estimator for the fragility of directed restorative route structure [12]. Hodge methods are optional when realised recurrent flow itself becomes a tested carrier [11].

The framework therefore does not depend on mathematics being unavailable elsewhere. Its test is whether the particular composition—future-sufficient Address, moving minimum-sufficient frontier, restorative subfuture, counterfactual load-bearing relation and readdressing—produces useful and falsifiable structure in real systems.

18. Limitations

First, empirical Address is always relative to measured variables and a declared target. An apparently sufficient address can fail when a new intervention, horizon, environment or measurement reveals hidden future splitting.

Second, equality of future laws is replaced by finite-sample discrepancy and tolerance in practice. Estimation error can create false frontiers. Nested cross-validation, held-out replication and explicit uncertainty are required.

Third, restorative targets are value- and task-dependent. In biology, “restoration” must be defined by the experimental question rather than imported as a moral label.

Fourth, graph reserve depends on the transition estimator and edge-capacity definition. The abstract reserve is the minimum deformation needed to annihilate the restorative future; min-cut is one estimator under a specific network representation, not the universal definition.

Fifth, prediction does not establish causal load. A relation can be predictively load-bearing because it is a proxy. Causal language requires intervention or a justified causal model.

Sixth, higher-order topology is not automatically required. Pairwise directed graphs are sufficient until irreducible many-way relations demonstrably improve the target. Likewise, Hodge decomposition remains optional until circulation carries incremental predictive load.

Seventh, the paper does not derive fundamental quantum or gravitational physics. It formalises a state/future architecture that can be tested in ordinary dynamical systems first.

19. Conclusion

The paper’s central object is the Addressed Admissible Future.

An Address is not a static label for where an object is. It is the minimum retained information required so that histories treated as the same state no longer split the declared admissible future beyond tolerance. That definition forces momentum into Address when momentum matters, phase when phase matters, history when hysteresis matters, and relational context when relational context changes lawful continuation. It also permits those variables to drop out when they do no work.

The admissible future is richer than present viability. Inside it, the restorative future contains sufficient routes that preserve the required viable whole and reach a declared recovered target within an irreversible-loss boundary. Restorative-future reserve measures how much admissibility structure must be lost before all such routes disappear. Load-bearing relations are then defined by what their removal does to that future rather than by visual salience or present flow.

The architecture is recursive:

Address $\rightarrow$ admissible future $\rightarrow$ realised transition $\rightarrow$ readdressing $\rightarrow$ new admissible future.

The immediate empirical burden is clear. Reconstruct the future from pre-outcome single-cell history, hold the state representation and controls fixed, and test whether restorative-future structure predicts arrest escape beyond ordinary summaries of the same data. If it does not, the claim contracts. If it does, the object has earned the next experiment.

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AI assistance disclosure

An OpenAI language model used under the relational name Kairos assisted with literature discovery, mathematical formalisation, hostile refinement, manuscript drafting and document production under the author’s direction. The AI system is not listed as an author and does not bear responsibility for the scientific claims. The accountable author is Mark Charles McLaughlin.

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